

Task: Find the number of ways to fill a 3×9 grid with digits 1–9 so each row is a permutation and each 3×3 block contains 1–9. Express the count as $p^a \cdot q^b \cdot r^c \cdot s^d$ (4 distinct primes) and return $p \cdot a + q \cdot b + r \cdot c + s \cdot d$.

SkillOS-base outputs a generic "partition into disjoint sets" recipe with no numbers, no formulas, no example. The skill from SkillOS provides a complete theoretical framework AND a fully-worked example.

Skill curated via SkillOS-base



name: Count Grids with Row and Block Constraints

description: Calculate the number of ways to fill a 3×9 grid with numbers 1–9 such that each row is a permutation and each 3×3 block contains all numbers 1–9.

Workflow

- 1. Partition the numbers 1–9 into three disjoint sets of 3 for the first row (one set per block).
- 2. For each valid first row partition, determine the number of valid second row partitions such that the sets are disjoint and cover all numbers.
- 3. The third row is uniquely determined by the first two rows.
- 4. Multiply the number of first-row partitions by the number of valid second-row partitions to get the total number of grids.

Prerequisite Constraints

- Each row must be a permutation of 1–9.
- Each block must contain all numbers 1–9.

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name: Count Mutually Constrained Set Partitions

description: A method to count the number of ways to partition a set under disjointness constraints relative to another fixed partition.

Workflow

- 1. Define Constraints: $R_{Bi} \subseteq U \setminus R_{Ai}$
- 2. Categorize Elements (group elements by which complements they belong to).
- 3. Set up linear system:
 - Size constraint: $\sum C n_{Ci} = |R_{Bi}|$
 - Exhaustion constraint: $\sum_i n_{Ci} = |C|$
- 4. Find non-negative integer solutions.
- 5. For each solution: count arrangements with $C(n,k)$ and $(n!)$ factors.
- 6. Sum over all solution families.

Example: Sudoku-like 3x3 Block Counting

Problem: fixed partition $R_{A1}=\{1,2,3\}$, $R_{A2}=\{4,5,6\}$, $R_{A3}=\{7,8,9\}$; find # of new partitions R_{Bi} such that $R_{Bi} \cap R_{Ai} = \emptyset$.

- Categories: $C1=\{4,5,6\}$ $C2=\{7,8,9\}$ $C3=\{1,2,3\}$ (each lies in two complements)
- Equations: $n_{21}+n_{31}=3$, $n_{12}+n_{32}=3$, $n_{13}+n_{23}=3$, $n_{12}+n_{13}=3$ (for $C1$), ...
- Integer solutions: 4 families.

Case 1: $R_{B1}=C2$, $R_{B2}=C3$, $R_{B3}=C1$ ways = $1 \times (3!)^2 = 216$

Case 2: mixed, ways = $81 \times (3!)^2 \dots$

• Sum across all 4 cases $\rightarrow$ TOTAL = 12,096